

A Simple Climate Model

CVL 2001



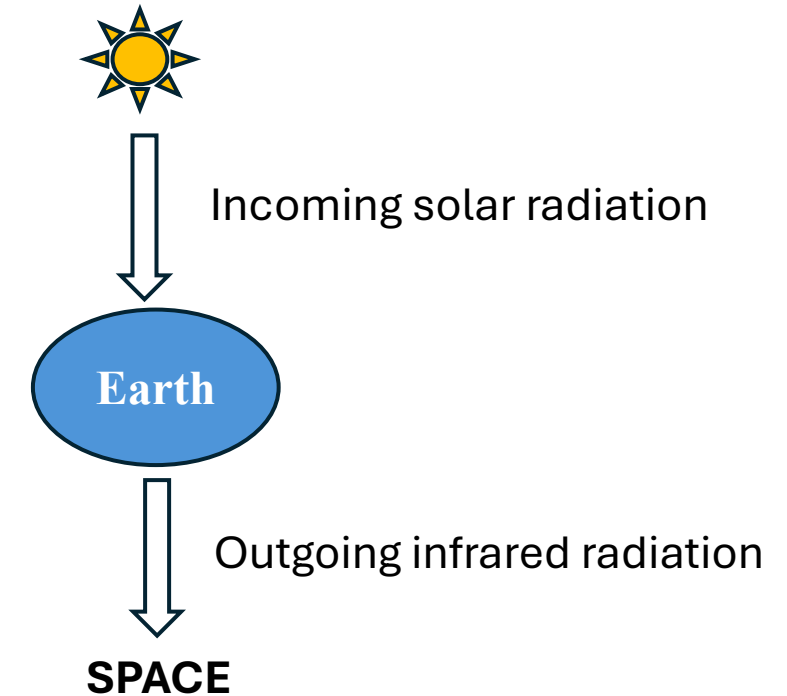
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Why does Earth have a temperature?

- Earth receives energy from the Sun and loses energy to space.
- If Earth receives **more energy than it loses**, it warms.
- If Earth loses **more energy than it receives**, it cools.
- At equilibrium:

Energy in = Energy out

- This is the **fundamental principle** behind the entire calculation.

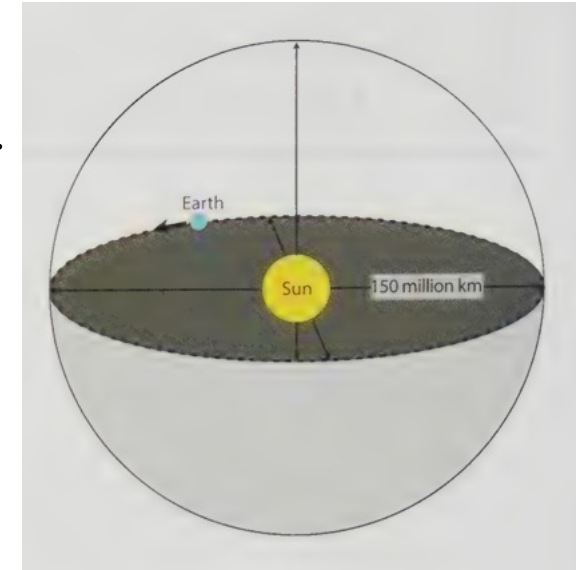


Where does Earth's incoming energy come from?

- Almost all of the energy driving Earth's climate system comes from the **Sun**.
- The Sun emits enormous amounts of energy in all directions.
- Imagine putting an enormous imaginary sphere around the Sun:
- As we move farther from the Sun, the same total solar energy is spread over a larger area.
- Therefore :

Solar radiation intensity decreases as $1/d^2$.

- This is why Venus receives much more solar radiation than Earth, while Mars receives less.
- The Earth's solar constant is approximately **$1360\text{--}1361 \text{ W m}^{-2}$**



What is the solar constant?

- The **solar constant**, S_0 , is approximately:

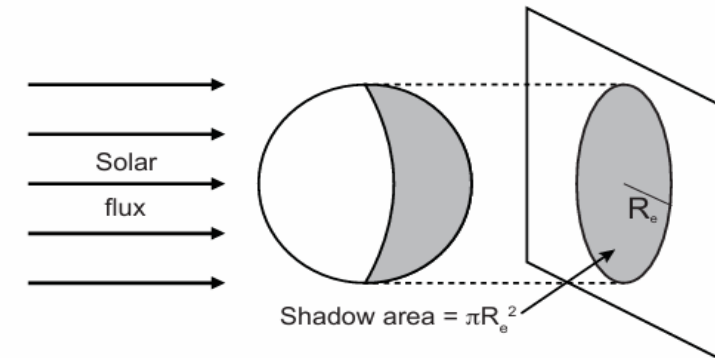
$$S_0 \approx 1361 \text{ W m}^{-2}$$

- This means that at the top of Earth's atmosphere, a surface oriented perpendicular to incoming sunlight receives roughly:
- **1361 joules every second per square metre.**
- **Earth does NOT absorb 1361 W/m² averaged over the whole planet.**
- Earth is spherical.
- Sunlight comes approximately from one direction.
- Earth intercepts sunlight over a **circular cross-sectional area**, not its entire spherical surface.
- The area facing the Sun is:

$$A_{\text{shadow}} = \pi R_e^2$$

- But the total surface area of Earth is:

$$A_{\text{Earth}} = 4\pi R_e^2$$



CONT...

➤ Therefore:

$$\pi R_e^2 / 4\pi R_e^2 = 1/4$$

➤ Hence Global average incoming solar flux = $S_0/4$

➤ So:

$$1361/4 \approx 340 \text{ W m}^{-2}$$

➤ Therefore, before considering reflection, Earth's **global-average incoming solar radiation** is approximately:

$$340 \text{ W m}^{-2}$$

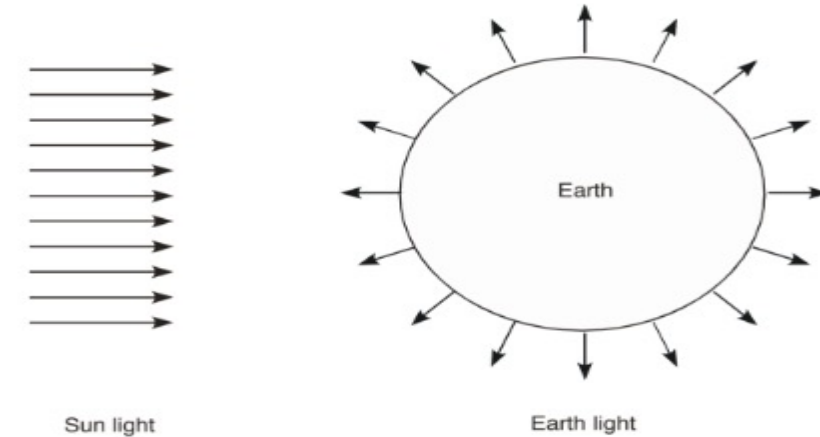
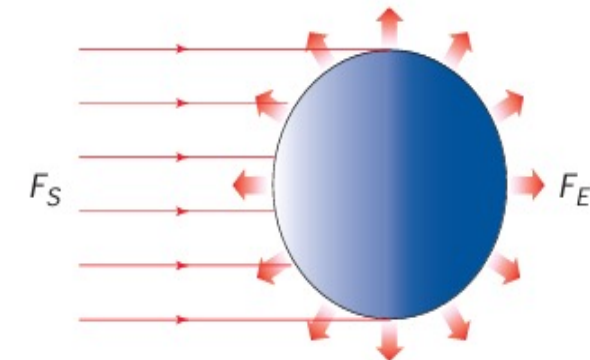


Fig : Earth's radiation balance showing incoming solar radiation intercepted over πR_e^2 and outgoing terrestrial radiation emitted over $4\pi R_e^2$



But Earth doesn't absorb all 340 W/m²?



No, then what happens ?



Earth Reflects back

Concept:

➤ **Albedo (α)** is the fraction of incoming solar radiation reflected back to space.

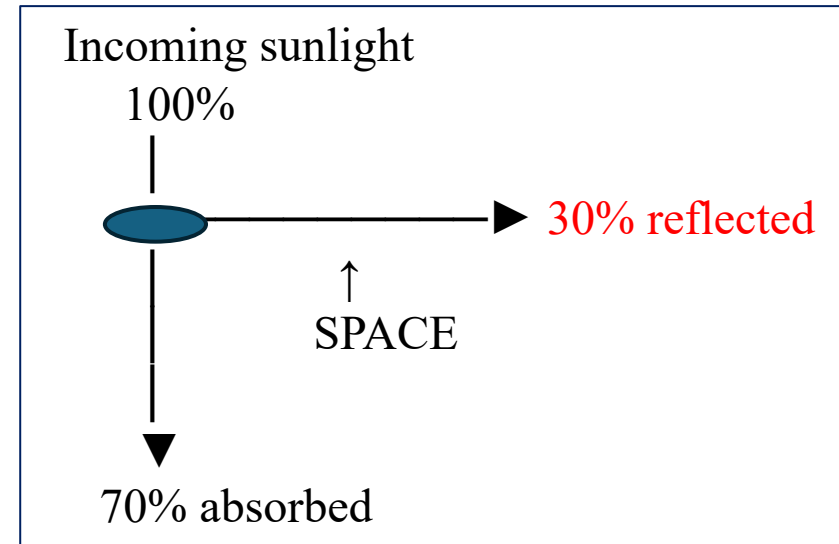
For Earth:

$$\alpha \approx 0.30$$

Therefore:

$$1 - \alpha = 1 - 0.30 = 0.70$$

➤ Only 70% is absorbed.



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Thus:

$$F_{\text{absorbed}} = S_0(1 - \alpha)/4$$

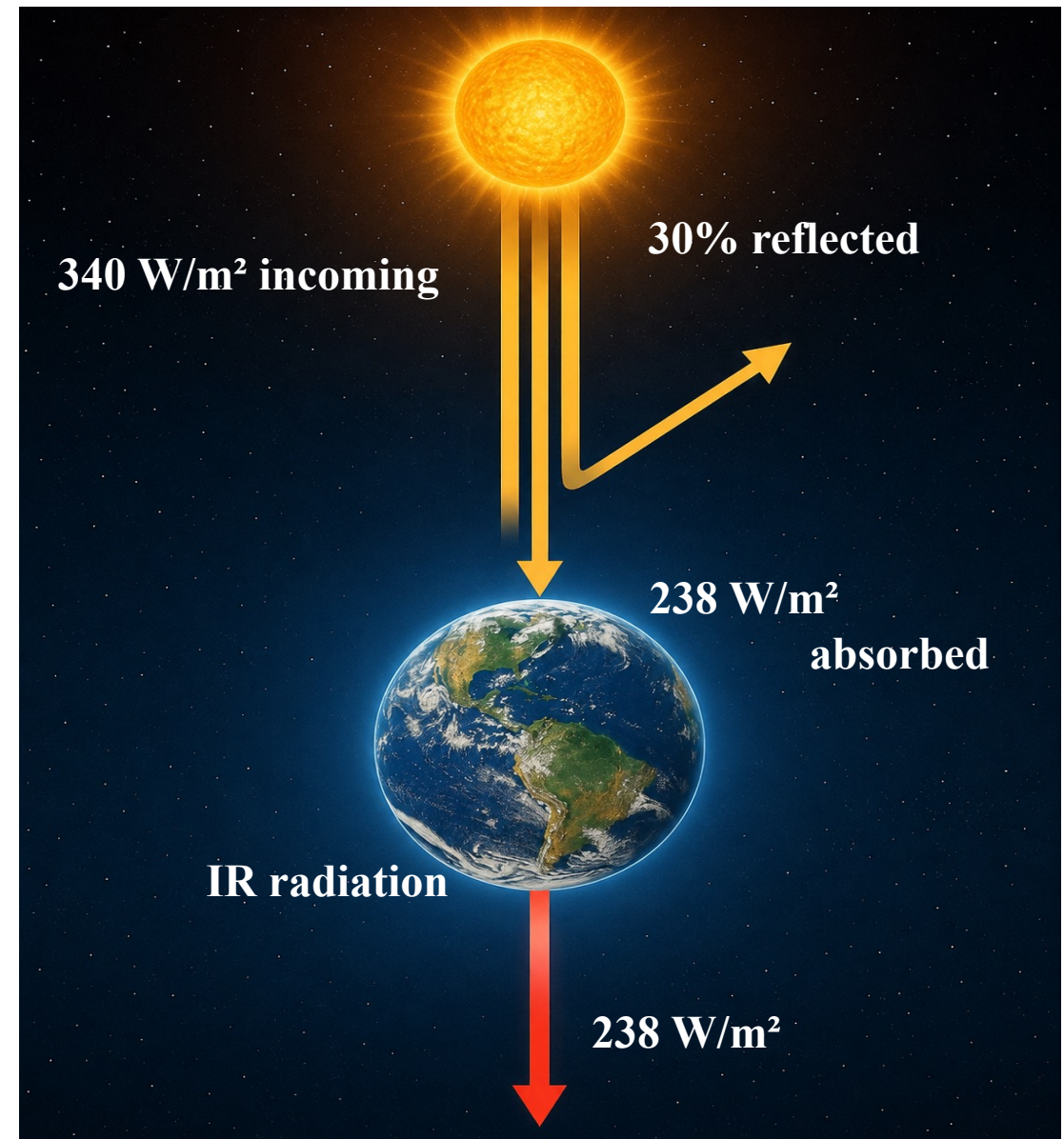
Using $S_0 = 1361 \text{ W m}^{-2}$

$$F_{\text{absorbed}} = 1361(0.7)/4$$

$$F_{\text{absorbed}} \approx 238 \text{ W m}^{-2}$$

- This is the famous **$\sim 240 \text{ W/m}^2$** number.
- The source similarly obtains about 238 W/m^2 for Earth's absorbed solar energy.
- But Earth cannot keep accumulating energy indefinitely.
- Therefore, eventually it must send approximately the same amount back into space.
- Therefore, at equilibrium:

$$\text{Energy}_{\text{in}} = \text{Energy}_{\text{out}} \quad (238 \text{ W m}^{-2} = 238 \text{ W m}^{-2})$$



Why does Earth emit infrared radiation?

WIEN'S DISPLACEMENT LAW

The wavelength at which a blackbody emits the maximum radiation is inversely proportional to its absolute temperature.

$$\lambda_{\max} T = b$$

(Wien's displacement law)

$\lambda_{\max}$ = wavelength of maximum emission (μm)

T = absolute temperature (K)

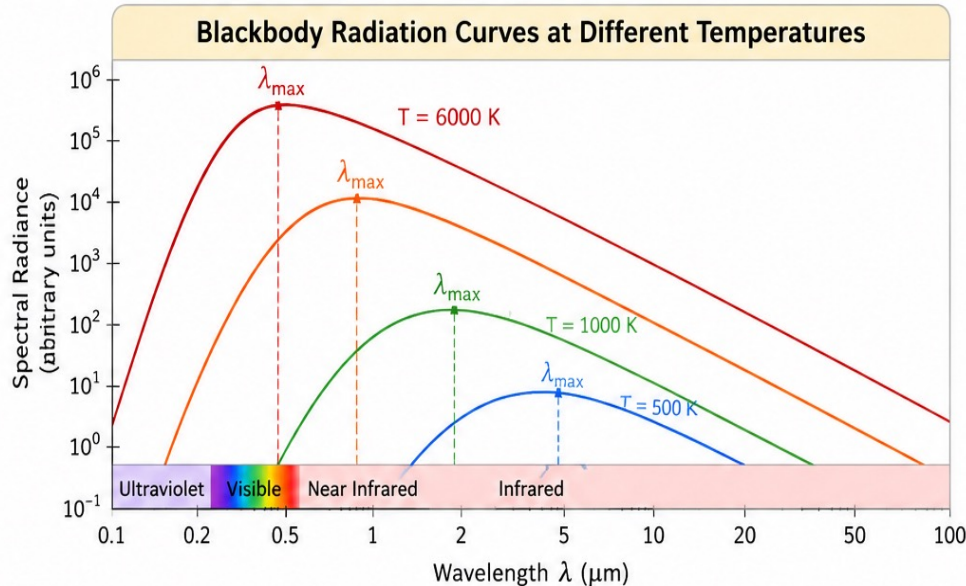
b = Wien's displacement constant

$$b = 2.898 \times 10^{-3} \text{ m} \cdot \text{K}$$

$$= 2898 \mu\text{m} \cdot \text{K}$$

Implications

- As temperature (T) increases, $\lambda_{\max}$ shifts to shorter wavelengths.
- Hotter objects emit peak radiation at shorter wavelengths.



Examples

Sun ($T \approx 5800 \text{ K}$)

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}}{5800} \text{ m}$$

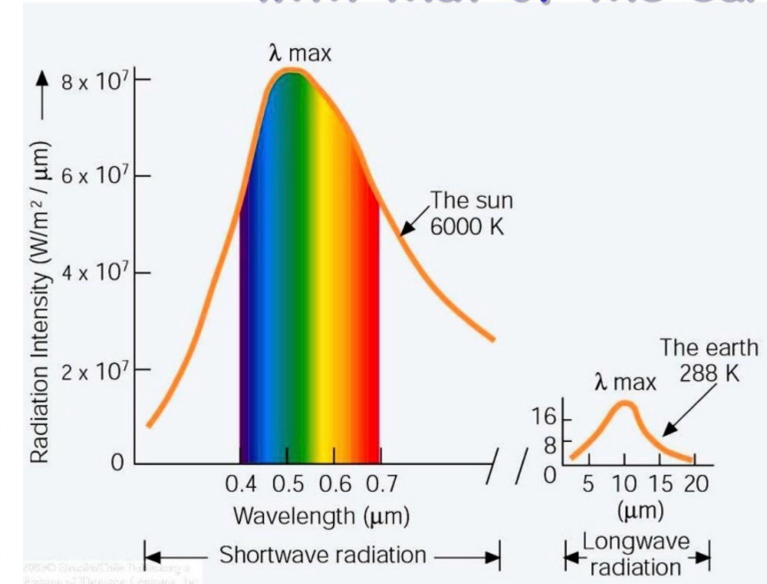
$\approx 0.50 \mu\text{m}$ (visible light)

Earth ($T \approx 288 \text{ K}$)

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}}{288} \text{ m}$$

$\approx 10.1 \mu\text{m}$ (infrared)

Spectrum of the sun compared with that of the earth



The hot sun radiates at shorter (visible) wavelengths that carry more energy

Energy absorbed by the cooler earth is then re-radiated at longer (thermal infrared) wavelengths

Sun → shortwave radiation (**extremely hot**)
Earth → longwave/infrared radiation (**much cooler**)

This difference is essential to the greenhouse effect.

Key Point: Higher temperature → shorter $\lambda_{\max}$ (blue). Lower temperature → longer $\lambda_{\max}$ (redder).

Earth with NO atmosphere

Imagine Earth has:

- no atmosphere
- no clouds
- no greenhouse gases
- no atmospheric absorption

Just a bare rocky Earth.(also known as bare rock model)

➤ Earth's temperature without atmosphere

Stefan–Boltzmann:

$$F = \sigma T^4$$

At equilibrium:

$$\sigma T_e^4 = 238$$

Therefore

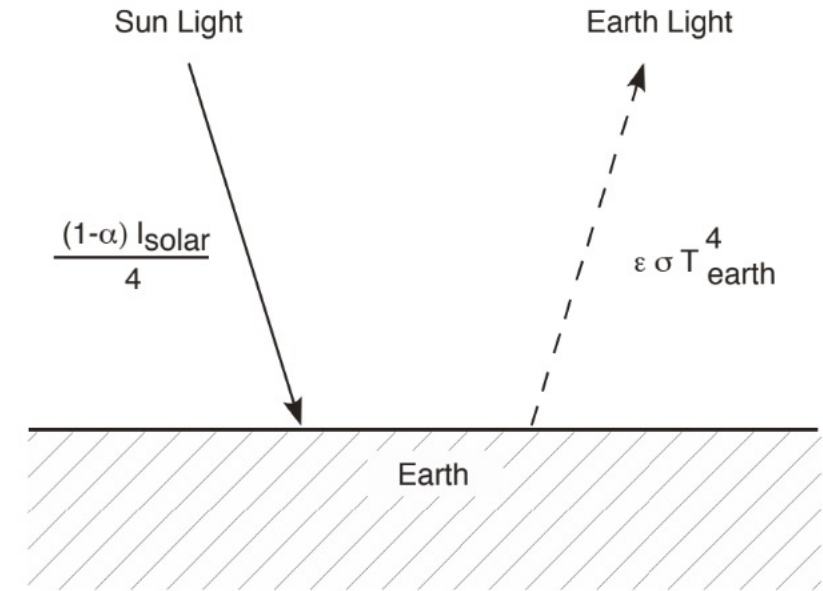
$$T_e = (238/\sigma)^{1/4}$$

where,

$\frac{P}{a}$ = Emitted power per unit area
(W/m²)

T = Absolute temperature (K)

σ = Stefan - Boltzmann constant
= $5.67 \times 10^{-8} \text{ W/m}^2/\text{K}^4$



or more generally $T_e = [S_0(1-\alpha)/4\sigma]^{1/4}$

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Putting in Earth's values:

$$T_e \approx 255\text{K}$$

Convert to Celsius:

$$T(^{\circ}\text{C}) = T(\text{K}) - 273.15 \quad 255 - 273.15 \approx -18^{\circ}\text{C}$$

So:

$$T_e \approx -18^{\circ}\text{C}$$

➤ But Earth's actual global-average surface temperature is roughly:

$$+15^{\circ}\text{C}$$

Difference:

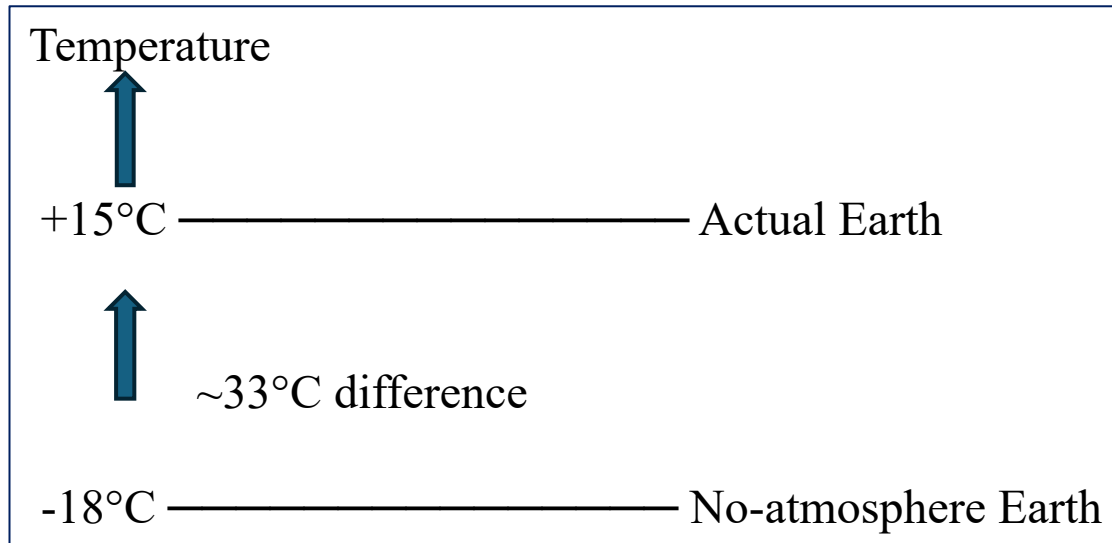
$$15 - (-18) = 33^{\circ}\text{C}$$

So approximately:

$$33^{\circ}\text{C}(\text{warmer})$$

What is responsible for this difference?

The greenhouse effect.





Now add an atmosphere

One-Layer model

- Imagine the atmosphere is just **one thin layer** surrounding Earth

But we make two assumptions.

- **Assumption 1**

The atmosphere is **transparent to sunlight**. So solar radiation passes through.

- **Assumption 2**

The atmosphere is **opaque/absorbing to Earth's infrared radiation**.

So Earth emits IR upward, but the atmosphere absorbs it

Note :

- If the atmosphere absorbs infrared energy, it also emits infrared radiation.
- And it emits in **both directions**.
- Atmosphere $\longrightarrow$ absorbs and re-emits IR

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The surface receives energy from **two sources**:

1. Solar radiation

$$238\text{W/m}^2$$

2. Downward atmospheric radiation

$$238\text{W/m}^2$$

Therefore, the total energy received by the surface is:

$$F_{\text{surface}} = 238 + 238 \quad F_{\text{surface}} = \mathbf{476\text{W/m}^2}$$

Surface Temperature in the One-Layer Model

Step 1: Stefan–Boltzmann Law

$$F = \sigma T_s^4$$

The surface must emit:

$$F = 476\text{W/m}^2$$

Therefore:

$$\sigma T_s^4 = 476$$

Step 2: Solve for surface temperature

$$T_s = (476/\sigma)^{1/4} \quad \mathbf{T_s \approx 303\text{K}}$$

Convert Kelvin to Celsius:

$$T_s = 303 - 273.15 \quad \mathbf{T_s \approx 30^\circ\text{C}}$$

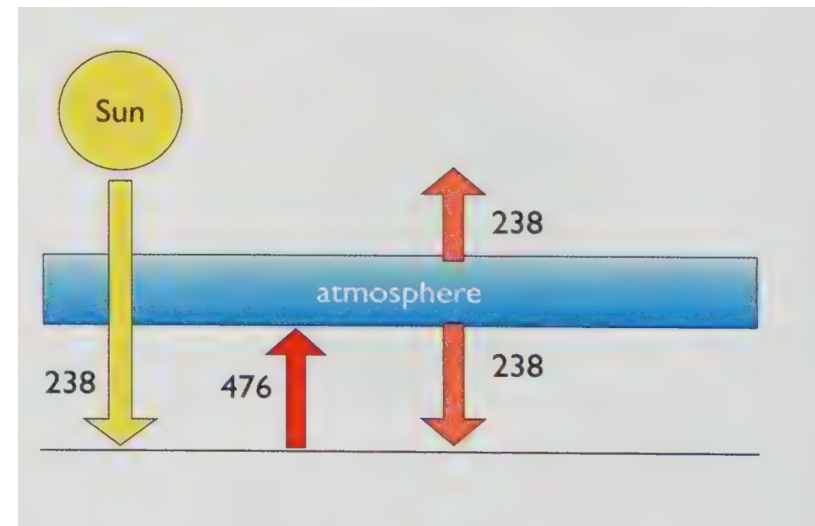


Fig : Energy flow in a one-layer atmosphere, transparent to visible radiation but absorbing infrared radiation, based on Earth's solar constant and albedo.



Effect of the Greenhouse Layer

Case	Surface Temperature
No atmosphere	255 K $\approx$ -18°C
One IR-absorbing layer	303 K $\approx$ 30°C

Temperature difference:

$$303-255=48\text{K}$$

Therefore:

$$\Delta T \approx 48^{\circ}\text{C}$$

Note

- The IR-absorbing atmosphere sends radiation back toward the surface, increasing the energy received by the surface.
The surface therefore warms until it can emit enough infrared radiation to restore energy balance.



Why Is It Called the Greenhouse Effect?

- The atmosphere acts **somewhat like a greenhouse** because it reduces the ease with which heat escapes.
- Greenhouse gases absorb **outgoing infrared (IR) radiation**.
- They then **re-emit IR radiation in both directions**, including downward toward the surface.
- This increases the energy received by the surface.
- As a result, the **equilibrium surface temperature increases**.

Note

- IR absorption ➡ IR re-emission ➡ more downward ➡ IR surface warming .

What Happens When We Add Two Atmospheric Layers?

- Add a second IR-absorbing atmospheric layer.
- Radiation is **absorbed and re-emitted multiple times**.
- Some radiation is emitted upward toward space.
- Some radiation is emitted downward toward the surface.
- The surface therefore receives **more downward IR radiation**.
- For the idealized two-layer model:

$$F_{\text{surface}} \approx 714 \text{ W/m}^2$$

Using the Stefan–Boltzmann law:

$$F = \sigma T_s^4$$

Gives:

$$T_s \approx 335 \text{ K } (62^\circ \text{C})$$

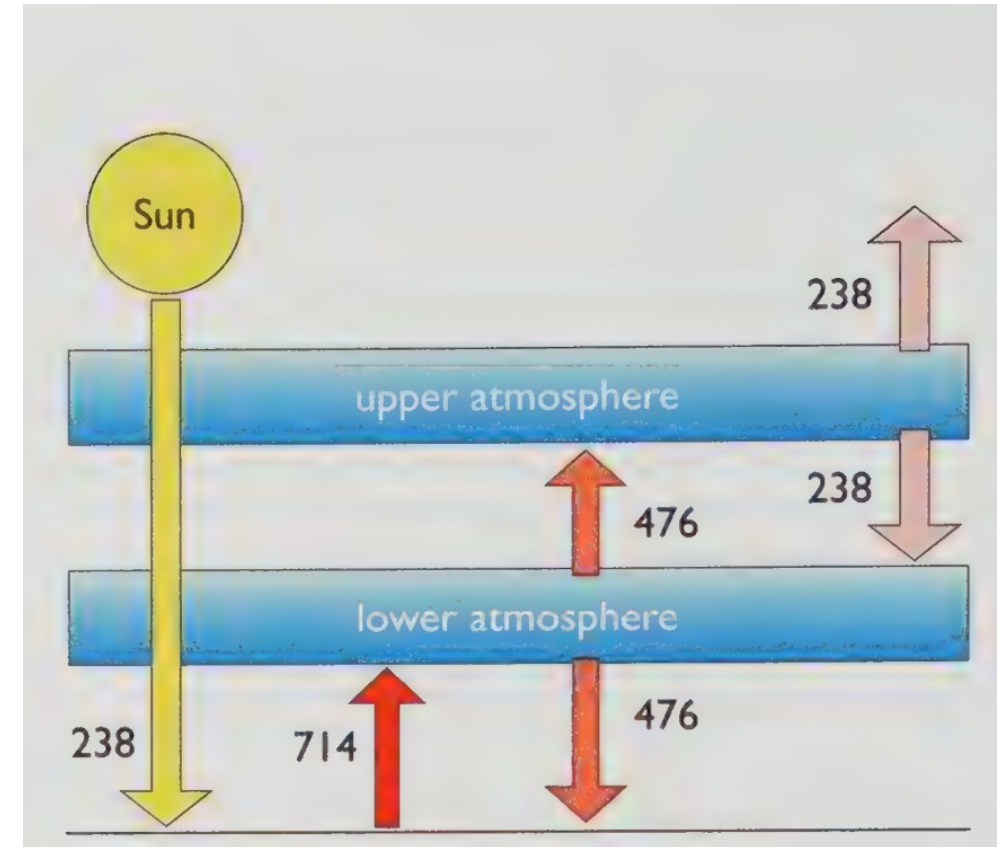
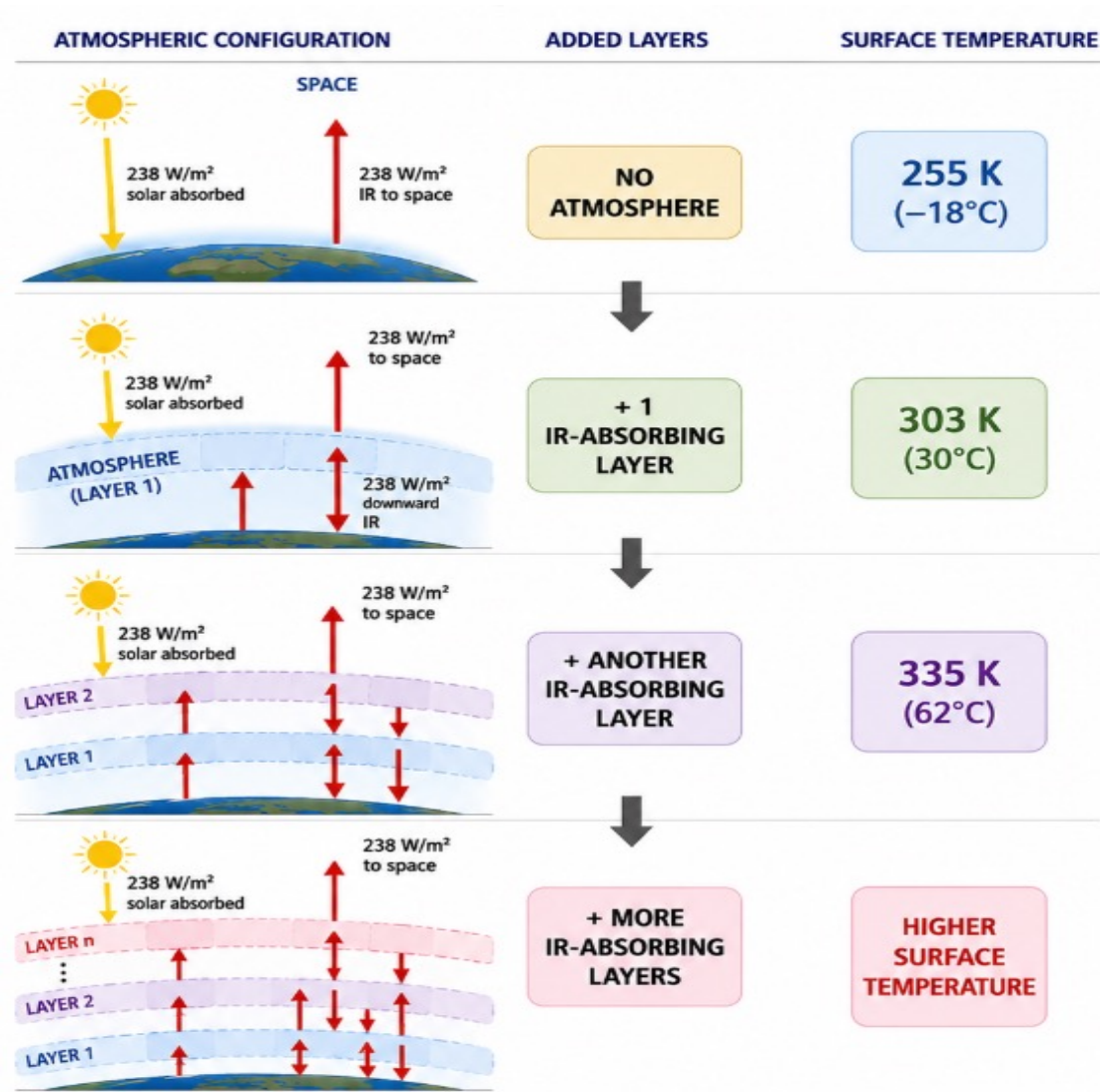


Fig: Energy flow in a two-layer atmosphere based on Earth's solar constant and albedo

ILLUSTRATIONS AND CONCEPT UNDERSTANDING



More IR-absorbing layers



More IR absorption



More IR re-emission



More downward IR flux



More energy to surface



Surface temperature (**Increases**)

An n-Layer Model

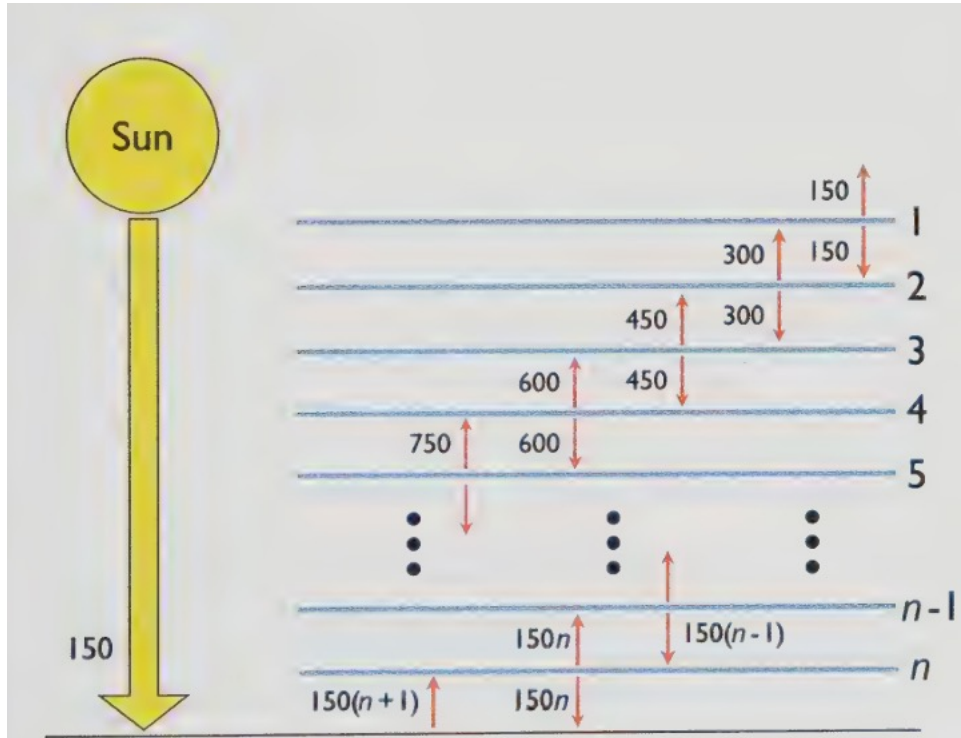


Fig :Energy flow in an n-layer atmosphere, showing radiative fluxes (W/m^2) for $S = 2000 \text{ W/m}^2$ and $\alpha = 0.7$.

➤ Example planet:

- Solar constant: $S=2000\text{Wm}^{-2}$
- Albedo: $\alpha=0.7$

➤ Incoming solar energy

- $F_{\text{in}}=S(1-\alpha)/4 = 2000(1-0.7)/4 = \mathbf{150\text{Wm}^{-2}}$

➤ Pattern

Layer 1: emits 150W/m^2 upward and downward.

Layer 2: emits 300W/m^2 upward and downward.

Layer 3: emits 450W/m^2 upward and downward.

Layer n: emits $150n\text{W/m}^2$ upward and downward.

Emission increases by 150Wm^{-2} for each additional layer



Surface Temperature of an n-Layer Planet

Surface Energy Balance

- The bottommost layer emits:

$$150 \text{ n } \text{Wm}^{-2}$$

downward toward the surface.

- The surface also receives:

$$150 \text{ Wm}^{-2}$$

directly from the Sun.

- Therefore, total energy received by the surface is:

$$F_{\text{surface}} = 150n + 150$$

$$F_{\text{surface}} = 150(n+1) \text{ Wm}^{-2}$$

For energy balance:

$$\sigma T_s^4 = 150(n+1)$$

Therefore:

$$T_s = [150(n+1)/\sigma]^{1/4}$$

- **General n-Layer Equation**

Since:

$$150 = S(1-\alpha) / 4$$

The general expression becomes:

$$T_s = [S(1-\alpha)(n+1)/4\sigma]^{1/4}$$

- **Parameters Controlling Surface Temperature :**

$$T_s = f(n, S, \sigma, \alpha)$$

Where:

n = number of atmospheric layers / greenhouse-gas absorption

S = solar constant

α = planetary albedo

σ = Stefan–Boltzmann constant (**$\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2} \text{ K}^{-4}$**)



Testing the n-Layer Model with Other Planets

Planet	Solar Constant (S) (W/m ²)	Albedo (α)	Energy In (W/m ²)	Surface Temp. (K)	Inferred (n)
Mercury	10,000	0.10	2,250	452	0.052
Venus	2,650	0.70	199	735	82
Earth	1,360	0.30	238	?	0.65
Mars	580	0.15	123	?	0.22

Planet	Inferred (n)	Greenhouse Effect	Interpretation
Mercury	≈ 0	Very weak	Essentially no atmosphere
Mars	0.22	Weak	Thin CO ₂ -rich atmosphere
Earth	0.65	Moderate	Stronger greenhouse effect than Mars
Venus	82	Massive	Extremely high surface temperature

Calculate surface temperature for Earth and Mars ?



Bonus question for exam



Key Takeaways for Lecture

1. Earth's climate temperature is constrained by energy balance: energy in = energy out.
2. Global-average absorbed solar radiation $\approx S_0(1-\alpha)/4 \approx 238 \text{ W m}^{-2}$.
3. A no-atmosphere Earth has an emission temperature $\approx 255 \text{ K}$ (-18°C).
4. An IR-absorbing atmosphere warms the surface by returning part of the outgoing IR downward.
5. In the idealized layer model, more IR-absorbing layers $\rightarrow$ higher surface temperature.
6. The model captures the essential physics but is not a detailed climate forecast.



Thank you